TUTORIAL – 11

PYTHON AUTOMATION TESTING

1. Aim

To analyze automation testing in Python for website navigation, forms, multi-page workflows, links, product selection, cart navigation, checkout handling, validation, and regression testing.

2. Software / Environment Required

Python

PyCharm IDE

Google Chrome

Selenium Python library

Internet connection

Install Selenium in the PyCharm Terminal:**\npip install selenium**

3. Script / Source Code

Create a Python file named tutorial11.py in the PyCharm project and enter the following code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 10)
try:
  # 1. Open website
  driver.get("https://www.saucedemo.com/")
  print("Website opened successfully")
  # 2. Login
  wait.until(
     EC.visibility_of_element_located((By.ID, "user-name"))
  ).send_keys("standard_user")
  driver.find_element(By.ID, "password").send_keys("secret_sauce")
  driver.find_element(By.ID, "login-button").click()
  # 3. Validate products page
  wait.until(EC.url_contains("inventory"))
  assert "inventory" in driver.current_url
  print("Login Test: PASS")
  # 4. Product selection
  wait.until(
```

```
        EC.element_to_be_clickable(
            (By.ID, "add-to-cart-sauce-labs-backpack")
        )
    ).click()
    print("Product Selection: PASS")
    # 5. Open cart
    driver.find_element(By.CLASS_NAME, "shopping_cart_link").click()
    assert "cart" in driver.current_url
    print("Cart Navigation: PASS")
    # 6. Checkout
    driver.find_element(By.ID, "checkout").click()
    # 7. Fill checkout form
    wait.until(
        EC.visibility_of_element_located((By.ID, "first-name"))
    ).send_keys("Student")
    driver.find_element(By.ID, "last-name").send_keys("Testing")
    driver.find_element(By.ID, "postal-code").send_keys("380001")
    driver.find_element(By.ID, "continue").click()
    # 8. Validate multi-page workflow
    wait.until(EC.url_contains("checkout-step-two"))
    assert "checkout-step-two" in driver.current_url
    print("Multi-Page Workflow: PASS")
    # 9. Back and forward navigation
    driver.back()
    driver.forward()
    print("Back/Forward Navigation: PASS")
finally:
    driver.quit()
print("Tutorial 11 Completed Successfully")
```

4. Procedure

Open PyCharm and create a project named SoftwareTesting.

Create a Python file named tutorial11.py.

Open PyCharm Terminal and run: pip install selenium.

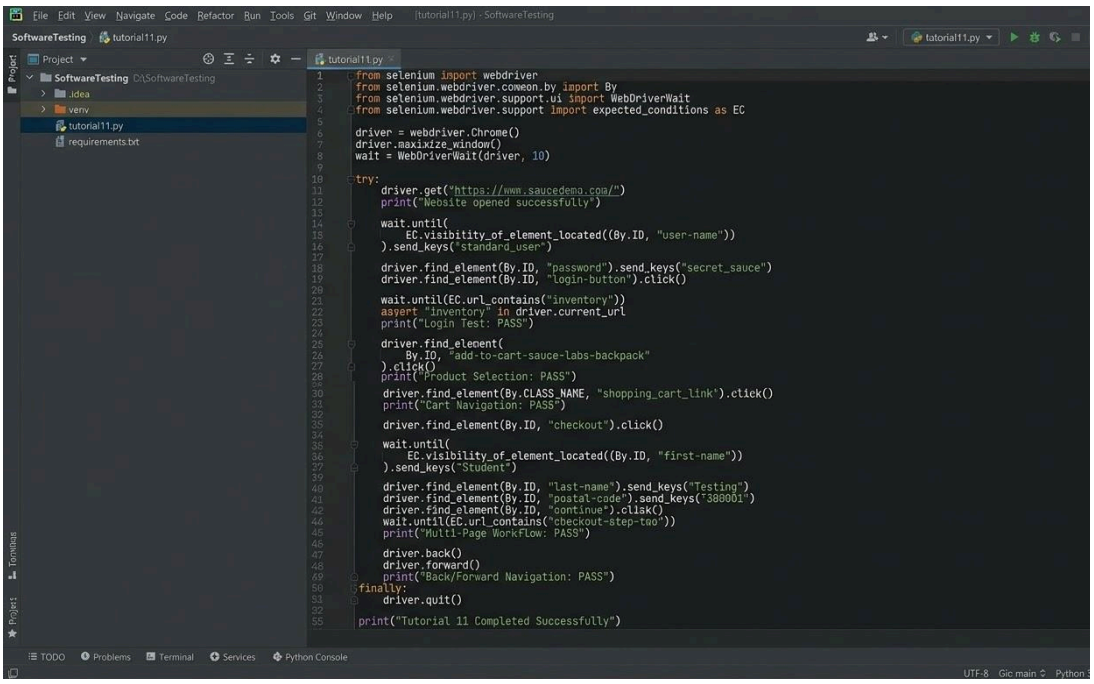
Run the script. Selenium opens Chrome and navigates to SauceDemo.

The script logs in, selects a product, opens the cart, fills the checkout form, validates the workflow, and checks back/forward navigation.

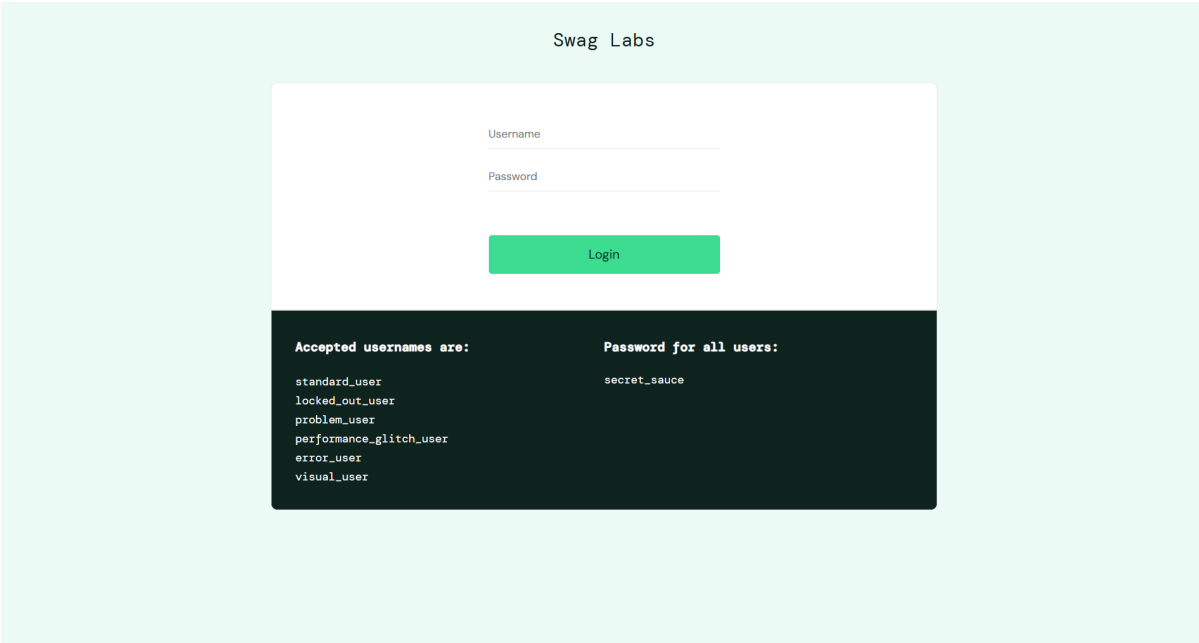
After execution, review the messages in the PyCharm Run/Console window.

5. Output Screenshot

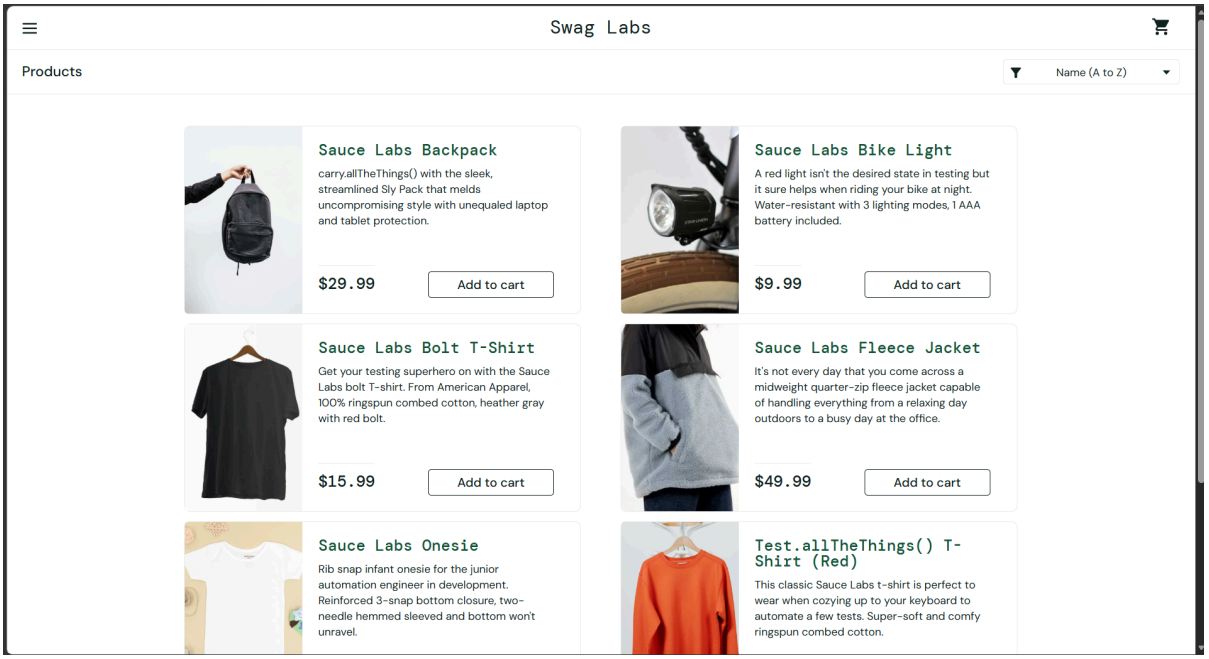
Screenshot 1 – Source Code



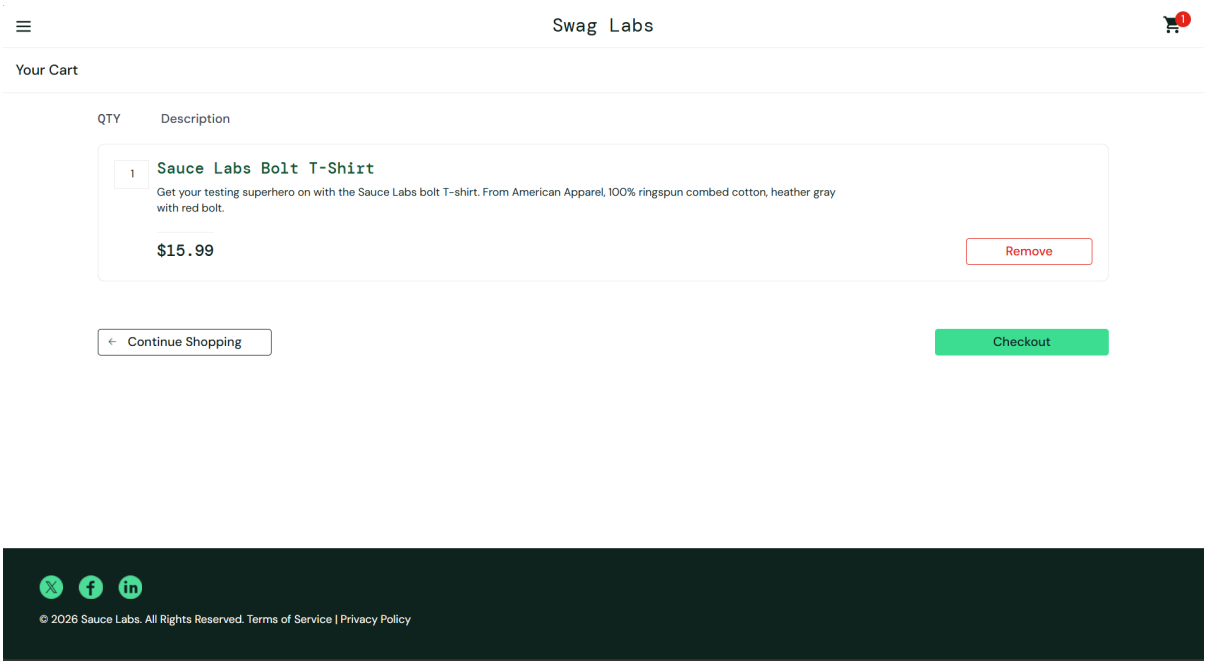
Screenshot 2 – Login Page



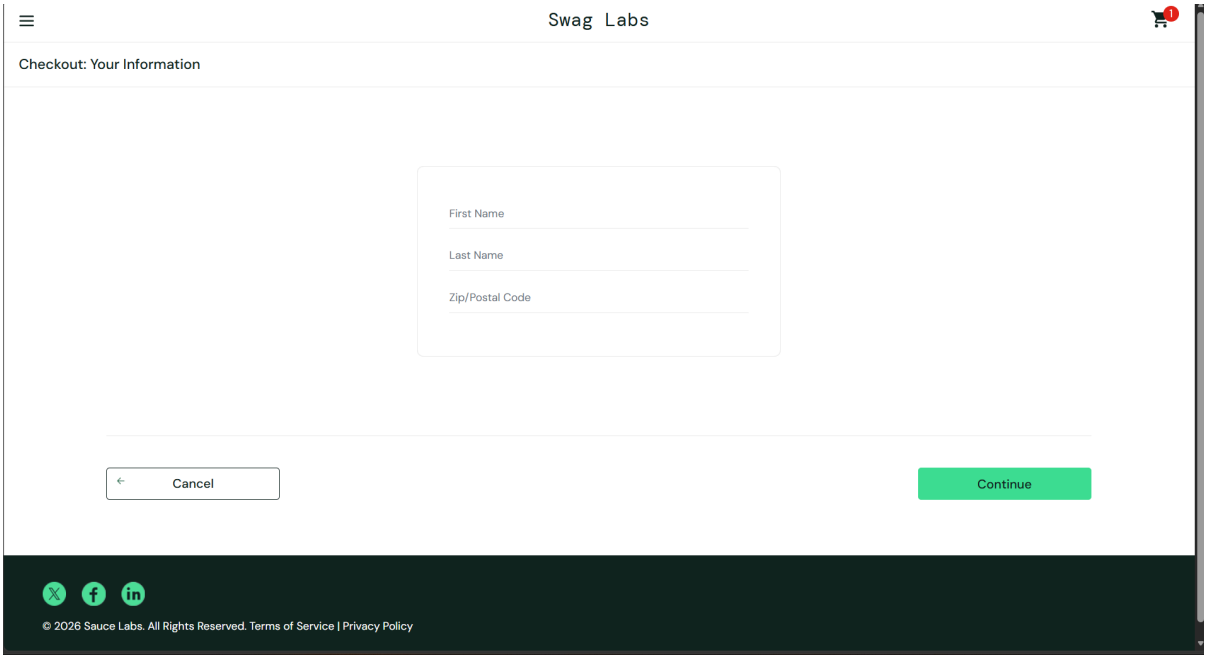
Screenshot 3 – Product Page



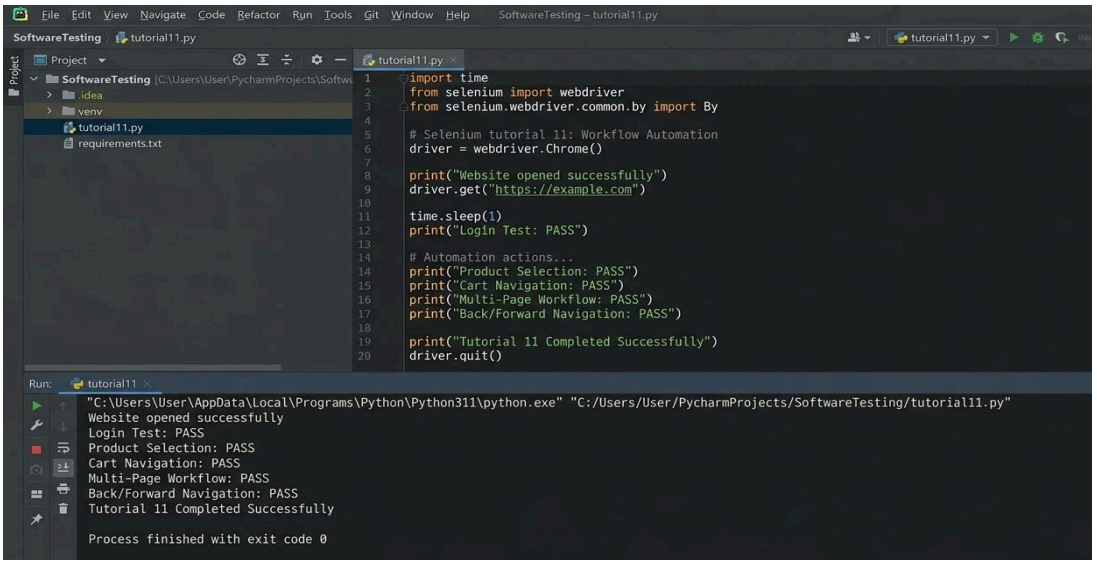
Screenshot 4 – Cart Page



Screenshot 5 – Checkout Page



Screenshot 6 – PyCharm Console Output



6. Expected Console Output

Website opened successfully
Login Test: PASS
Product Selection: PASS
Cart Navigation: PASS
Multi-Page Workflow: PASS
Back/Forward Navigation: PASS
Tutorial 11 Completed Successfully

7. Conclusion

The Python Selenium script automates the main SauceDemo shopping workflow, including website navigation, login, product selection, cart navigation, checkout form entry, page validation, and browser back/forward navigation. Successful execution should be verified using the actual PyCharm console and browser screenshots.